

Guhan Iyer

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EDUCATION

University of Waterloo

Sept. 2023 – Apr. 2028

Bachelor of Applied Science, Computer Engineering

Waterloo, Ontario

- Relevant Coursework: Real-Time Operating Systems, Algorithms & Data Structures, Computer Architecture, Digital Hardware Systems

PROFESSIONAL EXPERIENCE

Ford Motor Company

May 2026 – Aug. 2026

Software Development Intern

Waterloo, Ontario

- Developed **Python**-based security testing libraries to generate invalid ECU access tokens, enabling native testing of a daemon deployed across **all production ECUs**.
- Integrated security testing utilities into **Jenkins** testing pipelines, reducing validation setup time by **~15s** per test across **thousands** of daily CI runs.
- Migrated **30+** legacy tests to the new libraries and integrated with existing **Python** infrastructure, standardizing security workflows for **100+ engineers**.
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Nokia

Sept. 2025 – Dec. 2025

DSP Firmware Engineering Intern

Ottawa, Ontario

- Eliminated data corruption in a concurrent **C++ message-passing utility** by enforcing atomic packet writes, **resolving race conditions** and restoring data integrity.
- Built a **C++ 17 abstraction layer** to translate encoded chip configurations into typed structs, accelerating **hardware initialization** across **20+** operating modes.
- Implemented isolated **ADC/DAC initialization routines** and configuration paths, enabling targeted debugging of **individual ASIC modules** during hardware bring-up.
- Resolved **10+** major defects in the firmware SDK by analyzing trace logs and debugging with **GDB**, eliminating recurring system crashes.

Ford Motor Company

Jan. 2025 – Apr. 2025

Software Development Intern

Waterloo, Ontario

- Developed **Python**-based security testing libraries to generate invalid ECU access tokens, enabling native testing of a daemon deployed across **all production ECUs**.
- Integrated security testing utilities into **Jenkins** testing pipelines, reducing validation setup time by **~15s** per test across **thousands** of daily CI runs.
- Migrated **30+** legacy tests to the new libraries and integrated with existing **Python** infrastructure, standardizing security workflows for **100+ engineers**.

NCR Voyix

May 2024 – Aug. 2024

Software Engineering Intern

Waterloo, Ontario

- Designed a patch verification tool validating software deployments for **10,000+** devices across **10+** platforms.
- Integrated a **Python** query utility into a patch management system, reducing manual retrieval time by **60%**.

PROJECTS

Embedded Real-Time Kernel | C, STM32, ARM Cortex-M4

- Implemented a **preemptive Earliest Deadline First (EDF) scheduler** using **SysTick interrupts**, with task control blocks and context switching via **PendSV** and **SVC**.
- Built a custom **first-fit heap allocator** with freelist management, block coalescing, and runtime fragmentation tracking.

TECHNICAL SKILLS

Languages: C, C++, Assembly (ARM, RISC-V), Python, Java

Libraries & Tools: Valgrind, GDB, CMake, Make, Docker, Android Tools (ADB, fastboot)

Technologies & Protocols: Linux, QNX, FreeRTOS, TCP/IP, UART, I2C, gRPC, protobuf